

Keewoo Lee

✉ keewoole@gmail.com 🌐 <https://keewoolee.github.io>

OVERVIEW

I am a cryptography researcher in the Privacy Cluster at the Ethereum Foundation. Previously, I was a postdoctoral researcher at UC Berkeley, hosted by Prof. Sanjam Garg. I obtained my Ph.D. in Mathematical Sciences at Seoul National University, advised by Prof. Jung Hee Cheon. I am broadly interested in cryptography from theory to practice. Currently, my research focus is on cryptographic primitives for secure computation (e.g., homomorphic encryption, secure multiparty computation, zero-knowledge proof) and their applications (e.g., private database query), with emphasis on privacy in the Ethereum ecosystem.

EMPLOYMENT

Ethereum Foundation, Remote

- Scientific Advisor (Full-time) Dec 2025 – Present
 - Access Layer Cluster
- Scientific Advisor (Part-time) Feb 2025 – Nov 2025
 - Privacy Cluster

University of California, Berkeley, United States

- Postdoctoral Researcher Nov 2023 – Nov 2025
 - Host: Prof. Sanjam Garg

CryptoLab Inc., Republic of Korea

- Research Scientist (Freelancer), HealthcareAI Division Sep 2023 – Oct 2023
 - Focus: Privacy-preserving Machine Learning on Biomedical Data

EDUCATION

Seoul National University, Republic of Korea

- Ph.D. in Mathematical Sciences Sep 2017 – Aug 2023
 - Advisor: Prof. Jung Hee Cheon
 - Focus: Cryptography (Homomorphic Encryption, Secure Multiparty Computation, Lattice-based Cryptography)
 - Thesis: “A Study on Homomorphic Packing: Definitions, Constructions, and Limitations”
- B.S. in Mathematical Sciences Mar 2014 – Aug 2017

PUBLICATIONS

Authors are listed in alphabetical order by last name, unless an asterisk(*) is indicated.

CONFERENCE & JOURNAL

- [C21] K. Lee, “Barely Doubly-Efficient SimplePIR,” *Annual International Cryptology Conference (Crypto 2026)*
- [C20] Jaiden Fairoze, Sanjam Garg, K. Lee, Mingyuan Wang, “Bypassing Prompt Guards in Production with Controlled-Release Prompting,” *USENIX Security Symposium (USENIX Security 2026)*
- [C19] K. Lee, Yongdong Yeo, “SophOMR: Improved Oblivious Message Retrieval from SIMD-Aware Homomorphic Compression,” *USENIX Security Symposium (USENIX Security 2026)*
- [C18] Leo de Castro, Duhyeong Kim, Miran Kim, K. Lee, Seonhong Min, Yongsoo Song, “More Efficient Lattice-based OLE from Circuit-private Linear HE with Linear Overhead,” *Communications in Cryptology (IACR CiC)*, 2026
- [C17] *Mayank Rathee, K. Lee, Raluca Ada Popa, “Verifiable PIR with Small Client Storage,” *IEEE Symposium on Security and Privacy (S&P 2026)*
- [C16] *Grant Bosworth, K. Lee, Sunwoong Kim, “Leveraging FPGAs for Homomorphic Matrix-Vector Multiplication in Oblivious Message Retrieval,” *Asilomar Conference on Signals, Systems, and Computers (ACSSC 2025)*
- [C15] Leo de Castro, K. Lee, “VeriSimplePIR: Verifiability in SimplePIR at No Online Cost for Honest Servers,” *USENIX Security Symposium (USENIX Security 2024)*
- [C14] K. Lee, “Bit Security as Cost to Demonstrate Advantage,” *Communications in Cryptology (IACR CiC)*, 2024
 - Best Award, National Cryptography Contest 2022

- [C13] *Seoyoung Ko, K. Lee, Hyunhum Cho, Yoonjae Hwang, Huisu Jang, “Asynchronous Federated Learning with Directed Acyclic Graph-based Blockchain in Edge Computing: Overview, Design, and Challenges,” *Expert Systems with Applications*, 2023
- [C12] Jung Hee Cheon, K. Lee, “Limits of Polynomial Packings for $\mathbb{Z}_{p^k}$ and $\mathbb{F}_{p^k}$,” *Annual International Conference on the Theory and Applications of Cryptographic Techniques (Eurocrypt 2022)*
 - Best Award, National Cryptography Contest 2021
- [C11] Jung Hee Cheon, Dongwoo Kim, Duhyeong Kim, K. Lee, “On the Scaled Inverse of $(x^i - x^j)$ modulo Cyclotomic Polynomial of the form $\Phi_{p^s}(x)$ or $\Phi_{p^s q^t}(x)$,” *Journal of the Korean Mathematical Society*, 2022
- [C10] *Michael Cho, K. Lee, Sunwoong Kim, “HELPSE: Homomorphic Encryption-based Lightweight Password Strength Estimation in a Virtual Keyboard System,” *Great Lakes Symposium on VLSI (GLSVLSI 2022)*
- [C09] Jung Hee Cheon, Dongwoo Kim, K. Lee, “MHZ2k: MPC from HE over $\mathbb{Z}_{2^k}$ with New Packing, Simpler Reshare, and Better ZKP,” *Annual International Cryptology Conference (Crypto 2021)*
 - Excellence Award, National Cryptography Contest 2020
- [C08] *Wonkyung Jung, Eojin Lee, Sangpyo Kim, K. Lee, Namhoon Kim, Chohong Min, Jung Hee Cheon, Jung Ho Ahn, “Accelerating Fully Homomorphic Encryption Through Architecture-Centric Analysis and Optimization,” *IEEE Access*, 2021
- [C07] *Sunwoong Kim, K. Lee, Wonhee Cho, Yujin Nam, Jung Hee Cheon, Rob A. Rutenbar, “Hardware Architecture of a Number Theoretic Transform for a Bootstrappable RNS-based Homomorphic Encryption Scheme,” *2020 IEEE 28th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM 2020)*
- [C06] Jung Hee Cheon, Dongwoo Kim, Duhyeong Kim, Hun Hee Lee, K. Lee, “Numerical Methods for Comparison on Homomorphically Encrypted Numbers,” *International Conference on the Theory and Applications of Cryptology and Information Security (Asiacrypt 2019)*
 - Invited to *Journal of Cryptology* (Top 3 of 71 accepted papers among 307 submissions)
- [C05] *Sunwoong Kim, K. Lee, Wonhee Cho, Jung Hee Cheon, Rob A. Rutenbar, “FPGA-based Accelerators of Fully Pipelined Modular Multipliers for Homomorphic Encryption,” *2019 International Conference on ReConfigurable Computing and FPGAs (ReConFig 2019)*
- [C04] *Sungjoon Park, Minsu Kim, Seokjun Seo, Seungwan Hong, Kyoohyung Han, K. Lee, Jung Hee Cheon, Sun Kim, “A Secure SNP Panel Scheme using Homomorphically Encrypted K-mers without SNP Calling on the User Side,” *BMC Genomics*, 2019
- [C03] Jung Hee Cheon, Haejin Cho, Jaewook Jung, Joohee Lee, K. Lee, “Efficient Identity-Based Encryption from LWR,” *Annual International Conference on Information Security and Cryptology (ICISC 2019)*
- [C02] *Andrey Kim, Yongsoo Song, Miran Kim, K. Lee, Jung Hee Cheon, “Logistic Regression Model Training based on the Approximate Homomorphic Encryption,” *BMC Medical Genomics*, 2018
 - First Place Prize, iDASH Genomic Data Privacy and Security Protection Competition 2017
- [C01] Jung Hee Cheon, Jinhyuck Jeong, Joohee Lee, K. Lee, “Privacy-preserving Computations of Predictive Medical Models with Minimax Approximation and Non-adjacent Form,” *International Conference on Financial Cryptography and Data Security (WAHC 2017)*
 - Excellence Award, National Cryptography Contest 2016

BOOK CHAPTERS

- [B01] Laia Amorós, Syed Mahbub Hafiz, K. Lee, M. Caner Tol, “Gimme That Model!: A Trusted ML Model Trading Protocol,” In *Protecting Privacy through Homomorphic Encryption*, 2021

PREPRINTS

- [P03] Arka Rai Choudhuri, Sanjam Garg, K. Lee, Hart Montgomery, Guru Vamsi Policharla, Rohit Sinha, “A Cryptographic Framework for Proof of Personhood,” <https://eprint.iacr.org/2026/333>
- [P02] Arka Rai Choudhuri, Sanjam Garg, Matthew Gregoire, K. Lee, Mike Lodder, Hart Montgomery, Guru Vamsi Policharla, Jim Zhang, “Qurrency: a quantum-secure, private, and auditable platform for digital assets,” <https://eprint.iacr.org/2026/015>

[P01] Jung Hee Cheon, K. Lee, Jai Hyun Park, Yongdong Yeo, “Private Database Query with SIMD-Aware Homomorphic Compression,” <https://arxiv.org/abs/2408.17063>

HONORS & AWARDS

- Distinguished Reviewer Award, S&P 2026 May 2026
- Sejong Science Fellowship 2024–2025
National Research Foundation of Korea
≈\$50,000/year
- Doctoral Dissertation Award Apr 2024
Korean Mathematical Society
 - Best Award (\$1,000)
“A Study on Homomorphic Packing: Definitions, Constructions, and Limitations”
- Doctoral Dissertation Award Aug 2023
College of Natural Sciences, Seoul National University
 - Best Award (\$2,000)
“A Study on Homomorphic Packing: Definitions, Constructions, and Limitations”
- Global PhD Fellowship 2018–2023
National Research Foundation of Korea
Full Tuition and ≈\$15,000/year
 - Award for Top 10% of Global PhD Fellowship (\$4,000) May 2022
 - Award for Top 10% of Global PhD Fellowship (\$4,000) Mar 2020
- National Cryptography Contest
National Security Research Institute
 - Best Award (\$3,000) Oct 2022
“Bit Security as Cost to Observe Advantage”
 - Best Award (\$3,000) Oct 2021
“Limits of Polynomial Packings for $\mathbb{Z}_{p^k}$ and $\mathbb{F}_{p^k}$ ”
 - Excellence Award (\$2,000) Oct 2020
“MHZ2k: MPC from HE over $\mathbb{Z}_{2^k}$ ”
 - Excellence Award (\$1,500) Nov 2017
Problem-solving Track
 - Excellence Award (\$1,500) Nov 2016
“Privacy-Preserving Computation of Predictive Medical Models with Minimax Approximation”
- Best Paper Runner-up, Asiacrypt 2019 Dec 2019
International Association for Cryptologic Research
“Numerical Methods for Comparison on Homomorphically Encrypted Numbers”
Invited to *Journal of Cryptology* (Top 3 of 71 accepted papers among 307 submissions)
- First Place Prize, iDASH Genomic Data Privacy and Security Protection Competition Oct 2017
Track 3: Homomorphic Encryption (HME) based Logistic Regression Model Learning

INVITED TALKS

- Oblivious Compression for Homomorphic Encryption and Its Applications Feb 2025
EIMS-KMS International Workshop on Cryptography, Virtual
- Homomorphic Encryption: An Introduction to Secure Computation Nov 2024
ESL GCI Seminar Series @ Rochester Institute of Technology, New York, USA
- On the Bit Security of Cryptographic Primitives Oct 2022
2022 Korean Mathematical Society International Conference, Seoul, Korea
Invited Speaker of Focus Session on “Discrete Mathematics and Mathematics of Computer Science”
- Introduction to Secure Computation Mar 2022
BK21 Colloquium (Rookies Pitch) @ Seoul National University, Seoul, Korea
Invited as an Outstanding Graduate Student of Math@SNU

PRESENTATIONS

- Privacy & Scaling Explorations at Ethereum Foundation May 2025
Silicon Valley Private AI Forum (K-PAI), Santa Clara, CA
- Oblivious Message Retrieval for ZCash Jul 2024
MPC & FHE Primer @ EDCON2024, Tokyo, Japan
- VeriSimplePIR: Verifiability in SimplePIR at No Online Cost for Honest Servers Apr 2024
Bay Area Crypto Day

	▪ Homomorphic Packing: Constructions and Lower Bounds UC Berkeley Cryptography Seminar, Virtual	Feb 2023
	▪ Limits of Polynomial Packings for $\mathbb{Z}_{p^k}$ and $\mathbb{F}_{p^k}$ Eurocrypt 2022, Trondheim, Norway	May 2022
	2022 Korean Mathematical Society Spring Meeting, Virtual	Apr 2022
	▪ MHz2k: MPC from HE over $\mathbb{Z}_{2^k}$ with New Packing, Simpler Reshare, and Better ZKP Crypto 2021, Virtual	Aug 2021
	2020 Korean Mathematical Society Fall Meeting, Virtual	Oct 2020
	▪ Microsoft Private AI Bootcamp 2020 Korean Mathematical Society Spring Meeting, Virtual	Jul 2020
	▪ Numerical Methods for Comparison on Homomorphically Encrypted Numbers 2019 Korean Mathematical Society Spring Meeting	Apr 2019
	▪ Privacy-preserving Predictive Models with Minimax Approx. and Non-adjacent Form WAHC 2017, Sliema, Malta	Apr 2017
EXPERIENCES	▪ Visiting Student (Prof. Vinod Vaikuntanathan) MIT, Boston, Massachusetts, USA	Oct 2022–Dec 2022
	▪ Private AI Bootcamp Team Project: <i>Ensuring Trust when Trading ML Models</i> Microsoft Research, Redmond, Washington, USA	Dec 2019
SERVICES	▪ Program Committee Asiacrypt (2025, 2026), CCS (2026), S&P (2026), USENIX Security (2027), ISC (2025) • S&P 2026 Distinguished Reviewer Award ▪ Editorial Board IACR CiC (2025) ▪ Reviewer (Conferences) Crypto (2024, 2025, 2026), Eurocrypt (2025, 2026), Asiacrypt (2019, 2021, 2022, 2023, 2024), TCC (2024), PKC (2019), AsiaCCS (2023), CT-RSA (2019, 2020), PQCrypto (2020, 2023, 2024), ANTS (2020), FHE.org Workshop (2022), Mathcrypt Workshop (2023) ▪ Reviewer (Journals) Journal of Cryptology (JoC), Transactions on Dependable and Secure Computing (TDSC), Transactions on Information Theory (TIT), Designs, Codes and Cryptography (DCC)	

[Last update : 2026-06-25]